1. Binary Classifier : $\mathbb{R}^d \longrightarrow \{-1, 1\}$

- 분류 과정 : $x \rightarrow h \rightarrow y$
- 데이터셋 : $D_n = \{(x^{(1)}, y^{(1)}), \ldots, (x^{(n)}, y^{(n)})\}$
- 오차함수 : $\begin{cases} \text{training} : \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{h(x^{(i)}) \neq y^{(i)}\} \\ \text{test} : (\text{generalization}) \end{cases}$

---

2. Learning Algorithm.

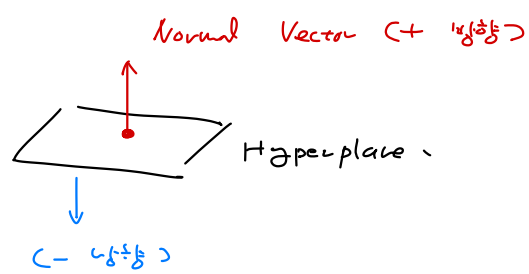
- Hypothesis Class : $\mathcal{H}$
- $D_n \longrightarrow$ learning $\text{alg}(\underbrace{\mathcal{H}}_{\text{restrict size (generalization)}}) \longrightarrow h$

---

3. Linear Classifiers

Note. two half-spaces

- 파라메터 $\begin{cases} \text{Normal vector} : \theta^d \in \mathbb{R}^d \\ \text{Scalar offset} : \theta \in \mathbb{R} \end{cases}$



- 분류 수식 $h(x; \theta, \theta_0) = \text{sign}(\theta^T x + \theta_0)$
$= \pm 1$

---

4. Random - Linear - Classifier

(i) k개의 무작위 $(\theta^{(j)}, \theta_0^{(j)})$

(ii) $j^* = \arg\min_{j \in \{1 \ldots k\}} \varepsilon_n(\theta^{(j)}, \theta_0^{(j)})$

---

5. Cross - Validation

(i) D into $D_1, D_2 \ldots D_k$ (k chunks)

(ii) train h_i on $D \setminus D_i$

(iii) test error의 평균 : $\frac{1}{k} \sum_{i=1}^{k} \varepsilon_i(h_i)$

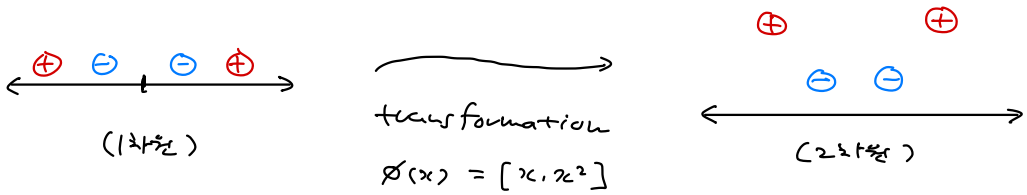
Note.

h의 성능이 아니라,

학습 알고리즘을 평가.

1. Linear Classifier의 한계와 Feature Transformation.

Linear Classifier는 Hypothesis Class가 제한적 (XOR dataset)



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2. Polynomial Basis

$\phi([x_1, x_2])$

$= [1, x_1, x_2, x_1^2, x_1x_2, x_2^2]$

order	$d=1$	in general
0	$[1]$	$[1]$
1	$[1, x]$	$[1, x_1 \dots x_d]$
2	$[1, x, x^2]$	$[1, x_1 \dots x_d, \overbrace{x_1^2, x_1x_2}^{\text{product of 2 diff dimension}}, \dots]$

---

3. Discrete Feature Encoding

- Numeric : 1.0/k, 2.0/k 1.0
- Thermometer code : $[1 \dots \underbrace{1}_{j\text{th}} 0 \dots \underbrace{0}_{k\text{th}}]$
- Factored code : [year, maker]
- One-hot code : $[0 \dots 0 \underbrace{1}_{j\text{th}} 0 \dots 0]$
- Binary code : length $\log(k)$

Note. A+ B+ AB+ O+
A− B− O−

(i) $[0, 0, 1, 0]$ $[1, 0]$
one hot one hot

(ii) $[\pm 1, \pm 1, \pm 1]$
A B ±

---

4. Other Features

- Texts
  ① fixed-length feature vector
  ② BoW (Bag of Words) $[0\ \underbrace{1}_{\text{등장하는 단어}}\ 0\ 0\ 1 \dots]$
- Numeric values
  ① $[-1, 1]$ scaling
  ② $\phi(x) = \dfrac{x - \bar{x}}{\sigma}$
  ③ higher order polynomial basis transformation

1. Notation

- feature x, label $y \in \{-1, 1\}$
- classifier $h_{\underbrace{w,b}_{\text{parameter}}}(x) = \text{sign}(w^T x + b) = \pm 1$

Note.

no estimation of $p(y=1)$

---

2. Functional Margin.

- 어떤 데이터 $\hat{\gamma}^{(i)} = y^{(i)}(w^T x^{(i)} + b) \underbrace{> 0}_{\text{정확한 예측의 경우}}$
- 데이터셋 $\hat{\gamma} = \min_{i=1,\dots,n} \gamma^{(i)}$
- Normalization $\|w\|_2 = 1$ ∵ otherwise $(w, b) \to (2w, 2b)$

  (i) $\text{sign}(w^T x + b)$ 은 유지됨.

  (ii) functional margin은 2배 증가.

---

3. Geometric Margin.

- 두점의 벡 $x^{(i)} - \gamma^{(i)} \frac{w}{\|w\|}$
- 결정경계 $w^T\left(x^{(i)} - \gamma^{(i)} \frac{w}{\|w\|}\right) + b = 0$

$$\gamma^{(i)} = \frac{w^T}{\|w\|} x^{(i)} + \frac{b}{\|w\|}$$

- Geometric Margin : $\gamma^{(i)} := y^{(i)}\left(\left(\frac{w}{\|w\|}\right)^T x^{(i)} + \left(\frac{b}{\|w\|}\right)\right)$

Note.

- $\|w\| = 1$ 이라면 func margin과 동일
- Scale invariance

  : w과 b를 rescale 해도 변하지 않음.

---

4. Optimal Margin Classifier

$$\max_{\gamma, w, b} \gamma \quad \text{s.t.} \quad y^{(i)}(w^T x^{(i)} + b) \geq \gamma, \quad \|w\| = 1$$

$\gamma = \frac{\hat{\gamma}}{\|w\|}$ $\Longrightarrow$

$$\max_{\hat{\gamma}, w, b} \frac{\hat{\gamma}}{\|w\|} \quad \text{s.t.} \quad y^{(i)}(w^T x^{(i)} + b) \geq \hat{\gamma}$$

$\hat{\gamma} = 1$ $\Longrightarrow$

$$\min_{w, b} \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y^{(i)}(w^T x^{(i)} + b) \geq 1$$